

Express.js — Day 7

HTTP Request Headers & Response Headers

Quick summary

Request headers carry information from the client to the server. **Response headers** carry information from the server back to the client. In Express, read request headers with **req.headers** and set response headers with **res.set()**.

1. Where headers fit in an HTTP message

A request has a method, URL, headers, and sometimes a body. A response has a status, headers, and often a body. Headers are metadata: they describe the request or response, rather than being the main data payload itself.

Request flow:

```
Postman / frontend → Request headers + body → Express server
```

Response flow:

```
Express server → Response status + headers + body → Postman / frontend
```

2. Reading request headers with req.headers

req.headers is an object containing the headers sent by the client. Header names are represented in lowercase in Express's Node.js request headers object, so use lowercase names when reading a specific header.

```
app.post('/users', (req, res) => {  
  console.log(req.headers);  
  res.json({ message: 'Request received' });  
});
```

To read one header, use its name as a key:

```
console.log(req.headers['content-type']);
```

If Postman sends JSON, the terminal can print:

```
application/json
```

This value comes from the request Postman sends. It describes the media type of the body.

3. Understanding Content-Type

Content-Type tells the recipient what kind of data is being sent in the message body. In our exercise, Postman was configured to send raw JSON, so the request included **Content-Type: application/json**.

Connection to Day 6: **express.json()** is middleware that parses incoming JSON bodies, making the parsed object available as **req.body**. The header describes the data type; the middleware performs the parsing.

Remember: Content-Type is a label describing the body. It does not itself parse the body.

4. Other request headers observed in Postman

Header	Meaning in this exercise
content-type	The body is being sent as application/json.
user-agent	Identifies the client software; Postman showed PostmanRuntime/7.53.0.
accept	Indicates which response media types the client says it can accept; /* is broad.
host	The host and port being requested; here localhost:3000.
content-length	The size of the request body in bytes, when supplied.

5. Setting response headers with res.set()

A server can include headers in the HTTP response. Express provides **res.set(name, value)** to set a response header.

```
app.post('/users', (req, res) => {
  res.set('X-App-Name', 'My Express App');
  res.json({ message: 'Request received' });
});
```

In this example, the server adds the custom response header **X-App-Name: My Express App**. The header is separate from the JSON response body.

The second experiment changed it to:

```
res.set('X-Developer', 'Bhavana');
```

Postman displayed **X-Developer: Bhavana** in the response Headers section. That confirmed the server had sent the custom header.

6. Finding headers in Postman

Request headers: click the Headers tab in the upper request configuration area. These are the headers Postman sends to Express.

Response headers: in the lower response panel, open the dropdown currently showing Body and select Headers. These are the headers Express sends back to Postman.

In the Day 7 screenshot, the response headers included **X-Powered-By: Express**, **X-Developer: Bhavana**, and **Content-Type: application/json; charset=utf-8**.

7. Complete practical example

```
const express = require('express');
const app = express();

app.use(express.json());

app.post('/users', (req, res) => {
  console.log(req.headers['content-type']);
  res.set('X-Developer', 'Bhavana');
  res.json({ message: 'Request received' });
});

app.listen(3000, () => {
  console.log('Server is listening on port 3000');
});
```

With Postman set to POST `http://localhost:3000/users` and Body → raw → JSON, the terminal prints `application/json`. The response Headers section shows X-Developer: Bhavana.

8. Request headers vs response headers

Question	Request headers	Response headers
Direction	Client → server	Server → client
Read / set in Express	<code>req.headers</code>	<code>res.set(name, value)</code>
Example	Content-Type: <code>application/json</code>	X-Developer: Bhavana
Where to inspect	Postman request Headers tab	Postman response Headers section

9. Day 7 knowledge check

Question 1: What does `req.headers['content-type']` read?

Answer: The Content-Type header sent by the client; in our Postman request it was `application/json`.

Question 2: What does `res.set('X-Developer', 'Bhavana')` do?

Answer: It adds the X-Developer response header with the value Bhavana.

Question 3: Where did you see X-Developer in Postman?

Answer: In the response Headers section, not in the response JSON body.

10. Long-term memory shortcut

`req.headers` → read headers that arrive with the request.

`res.set(name, value)` → set a header that will be sent with the response.

Content-Type: `application/json` describes the body being sent. `express.json()` parses incoming JSON so it can be accessed through `req.body`.

Final flow: Postman request headers + body → Express route → response headers + body → Postman

Express.js Day 7 — COMPLETE ✓